Unified Modeling Language (UML)

Introduction

- UML is a standard visual **language** used to **model**, **design**, and **document** software systems.
- Helps **developers**, **analysts**, and **stakeholders** **understand** system **requirements** and **ideas** clearly using **diagrams**.
- UML **improves planning**, **reduces** errors, and **provides** clear **documentation** throughout the software development process.

Who Benefits from UML?

Developers

Guidance for implementation and clarity.

Analysts

Capture and refine system requirements.

Designers

Structure system architecture effectively.

Stakeholders

Understand system behavior clearly.

When to Apply UML

Requirements Phase

During gathering and analysis phases.

Design Phase

For system architecture and components.

Documentation

As a comprehensive project documentation tool.

Communication

Communicate effectively with stakeholders.

UML Diagram Types

Structure Diagrams

Illustrate static aspects of a system.

Behavior Diagrams

Show dynamic system behavior.

Interaction Diagrams

Focus on control and data flow.

Structural Diagrams

Focus on "What"

Components and their relationships.

Static Aspects

Represent parts that do not change.

Class Diagrams

Example: model classes, attributes, operations.

Behavioral Diagrams

Focus on "How"

Processes and interactions of the system.

Dynamic Aspects

Represent parts that change over time.

Use Case Example

Illustrates user interactions with the system.

Interaction Diagrams: Who Talks

Subset of Behavior

Specialized for object communication.

Message Order

Focuses on communication sequence between objects.

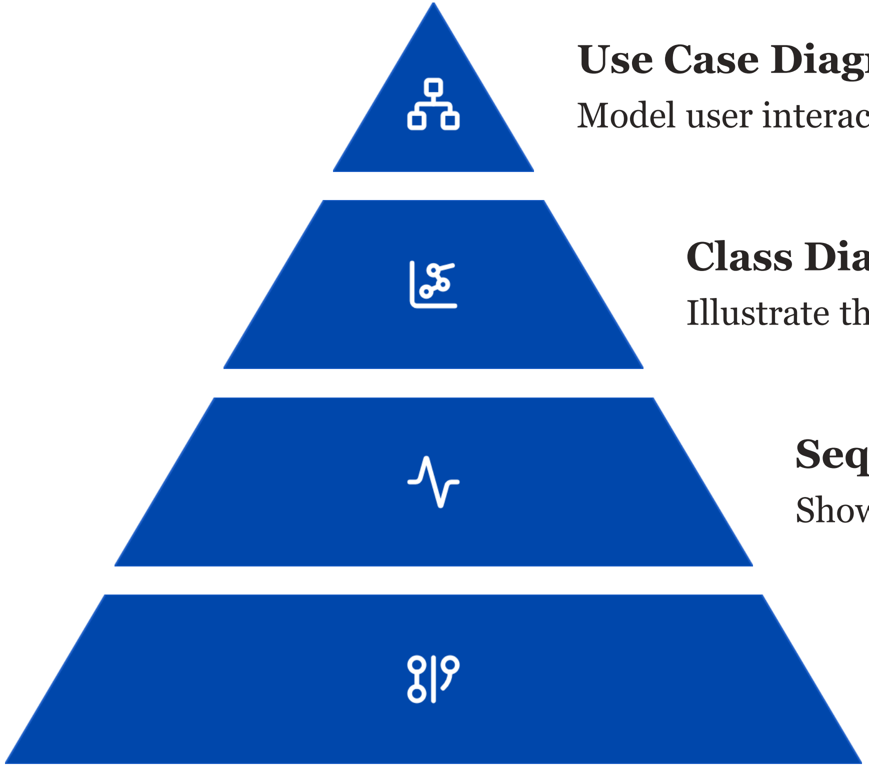
Sequence Diagram

Shows interactions in chronological order.

Communication Diagram

Illustrates object message exchanges.

Essential UML Diagrams



Use Case Diagrams

Model user interactions with the system.

Class Diagrams

Illustrate the static structure of the system.

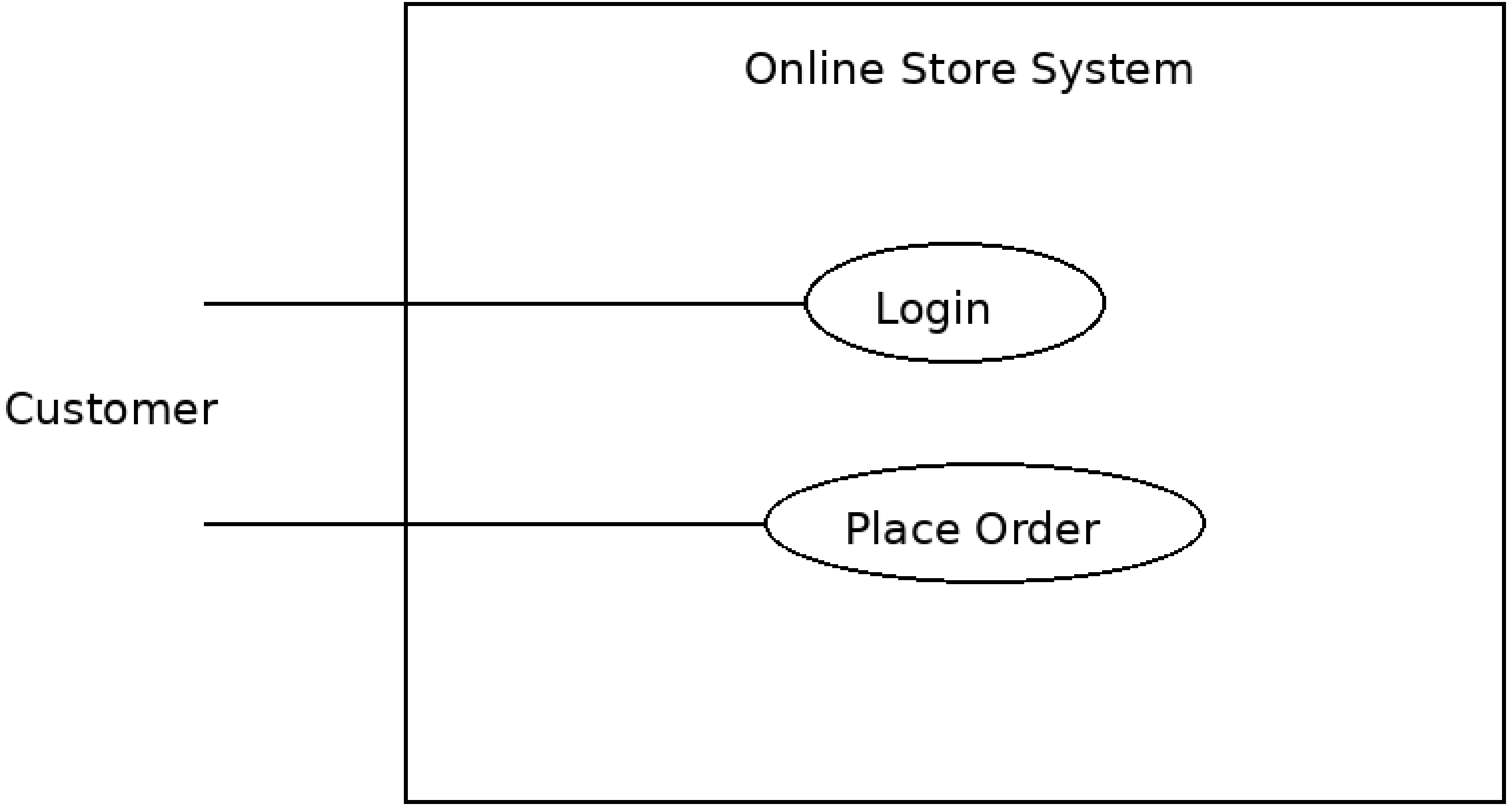
Sequence Diagrams

Show object interactions in time-ordered sequence.

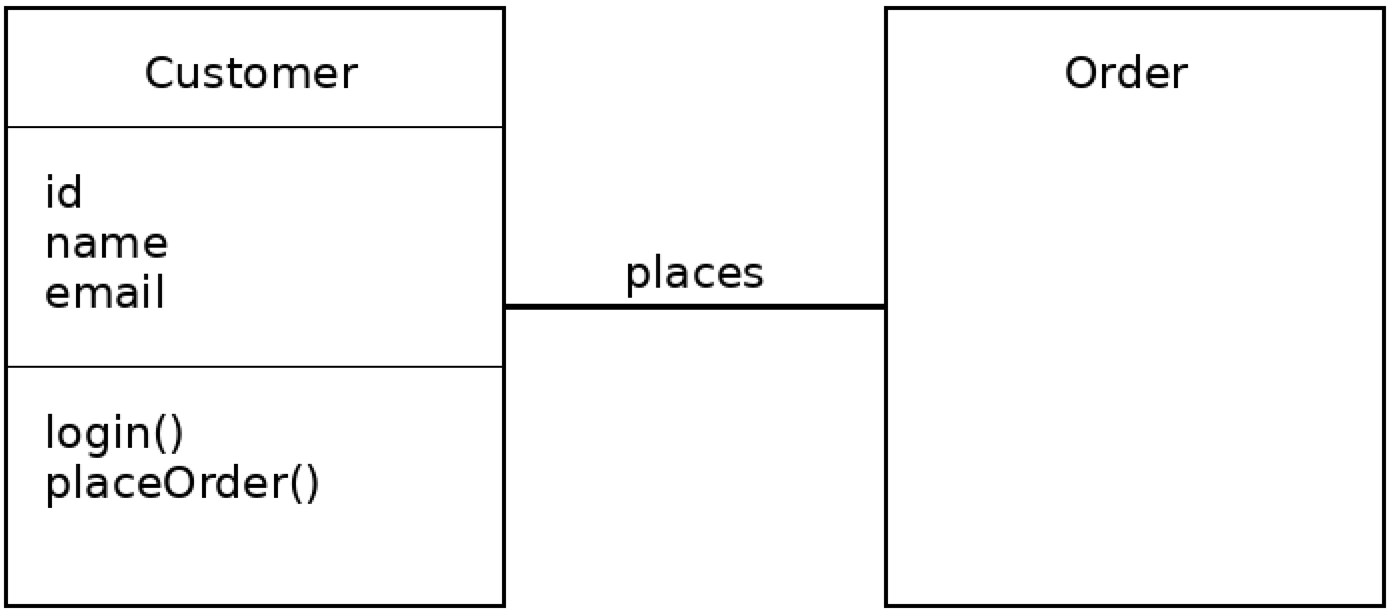
Activity Diagrams

Depict the flow of control in a system.

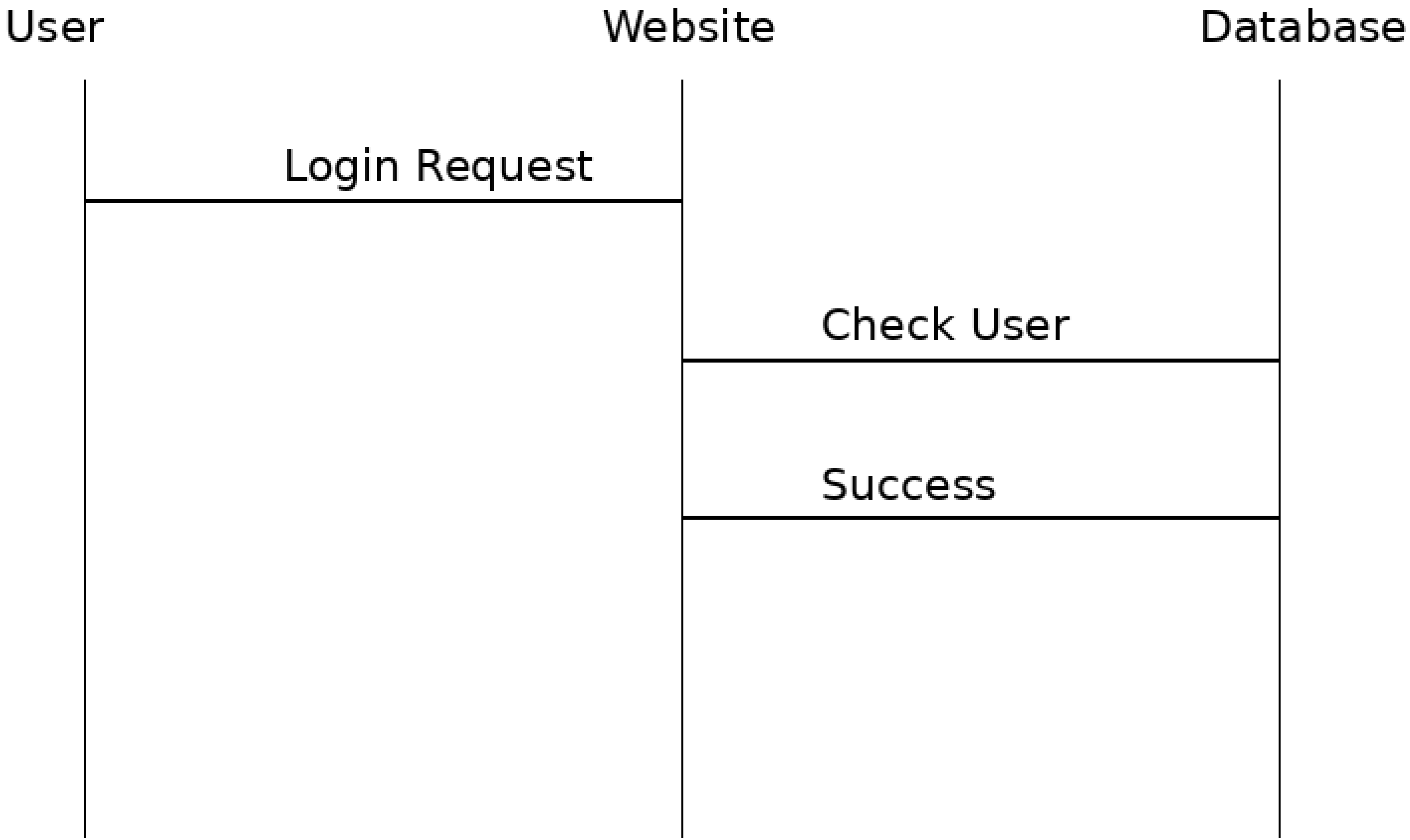
Use Case Diagram



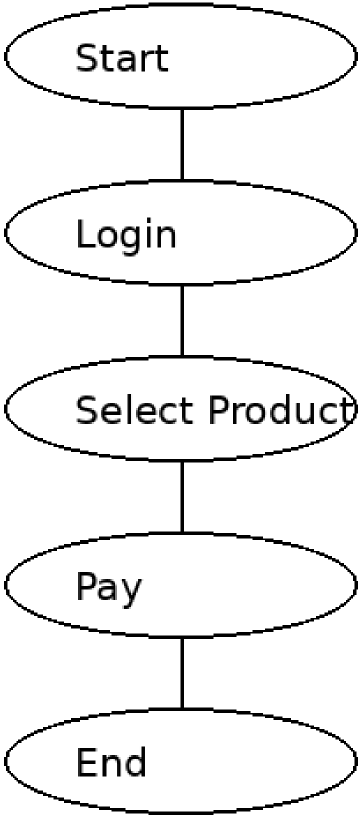
Class Diagram



Sequence Diagram



Activity Diagram



Assignment

- Explain the purpose and key components of a UML Use Case Diagram using a library management system as an example.
- Discuss the role of a UML Class Diagram in representing the static structure of a student registration system.
- Explain how a UML Sequence Diagram models interactions between objects in an e-commerce order processing system.
- Describe the importance of a UML Activity Diagram in modeling workflows, using an online course enrollment process as an example.